

CUSTOMER CHURN ANALYSIS AND PREDICTION

Machine Learning Internship Report

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ABSTRACT

This project focuses on predicting customer churn in a telecommunications company using machine learning techniques. The Telco Customer Churn dataset containing 7043 records was analyzed and processed. Data preprocessing, feature selection, model selection, model training, and evaluation were performed. A Random Forest Classifier achieved an accuracy of 79.21%, demonstrating its effectiveness in predicting customer churn.

INTRODUCTION

Customer churn refers to customers discontinuing a company's services. Predicting churn enables organizations to proactively retain customers and reduce revenue loss.

OBJECTIVES

- Analyze customer churn behavior
- Preprocess and clean the dataset
- Select important features
- Train a machine learning model
- Evaluate model performance

DATASET DESCRIPTION

Dataset: Telco Customer Churn Dataset

Records: 7043

Features: 21

Target Variable: Churn

TASK 1 – DATA PREPARATION

Removed customerID, handled missing values, converted TotalCharges to numeric format, and encoded categorical variables using One-Hot Encoding.

```
[10] df.info()
```

```
... <class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   str
1   gender                7043 non-null   str
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   str
4   Dependents            7043 non-null   str
5   tenure               7043 non-null   int64
6   PhoneService          7043 non-null   str
7   MultipleLines         7043 non-null   str
8   InternetService       7043 non-null   str
9   OnlineSecurity        7043 non-null   str
10  OnlineBackup          7043 non-null   str
11  DeviceProtection      7043 non-null   str
12  TechSupport           7043 non-null   str
13  StreamingTV           7043 non-null   str
14  StreamingMovies       7043 non-null   str
15  Contract              7043 non-null   str
16  PaperlessBilling      7043 non-null   str
17  PaymentMethod         7043 non-null   str
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   str
20  Churn                 7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

```
df.isnull().sum()

*** customerID      0
    gender          0
    SeniorCitizen   0
    Partner         0
    Dependents      0
    tenure          0
    PhoneService    0
    MultipleLines    0
    InternetService  0
    OnlineSecurity   0
    OnlineBackup     0
    DeviceProtection 0
    TechSupport      0
    StreamingTV      0
    StreamingMovies  0
    Contract         0
    PaperlessBilling 0
    PaymentMethod    0
    MonthlyCharges   0
    TotalCharges     0
    Churn            0
    dtype: int64
```

df = pd.get_dummies(df, drop_first=True)

df.head()

																				False
SeniorCitizen	tenure	MonthlyCharges	TotalCharges	gender_Male	Partner_Yes	Dependents_Yes	PhoneService_Yes	MultipleLines_No phone service	MultipleLines_Yes	StreamingTV_Yes	StreamingMovies_No Internet service	StreamingMovies_Yes	Contract_One year	Contract_Two year	PaperlessBilling_Yes	PaymentMethod_Credit card (nonsecure)	PaymentMethod_Electronic check	PaymentMethod_Mail check	Churn_Yes	
0	0	1	29.85	False	True	False	False	True	False	—	False	False	False	False	True	False	True	False	False	
1	0	36	56.95	True	False	False	True	False	False	—	False	False	True	False	False	False	False	True	False	
2	0	2	59.85	True	False	False	True	False	False	—	False	False	False	False	True	False	False	True	True	
3	0	40	43.90	True	False	False	False	True	False	—	False	False	True	False	False	False	False	False	False	
4	0	2	70.90	True	False	False	False	True	False	—	False	False	False	False	True	False	True	False	True	

TASK 2 – DATA SPLITTING

The dataset was split into 80% training data and 20% testing data for unbiased evaluation.

```
print("Training Features:", X_train.shape)
print("Testing Features:", X_test.shape)

print("Training Target:", y_train.shape)
print("Testing Target:", y_test.shape)
```

[7] ✓ 0.0s

```
... Training Features: (5634, 30)
Testing Features: (1409, 30)
Training Target: (5634,)
Testing Target: (1409,)
```

```

X_train.to_csv(
    r"C:\Users\dell\OneDrive\Desktop\Customer_Churn_Project\Outputs\X_train.csv",
    index=False
)

X_test.to_csv(
    r"C:\Users\dell\OneDrive\Desktop\Customer_Churn_Project\Outputs\X_test.csv",
    index=False
)

y_train.to_csv(
    r"C:\Users\dell\OneDrive\Desktop\Customer_Churn_Project\Outputs\y_train.csv",
    index=False
)

y_test.to_csv(
    r"C:\Users\dell\OneDrive\Desktop\Customer_Churn_Project\Outputs\y_test.csv",
    index=False
)

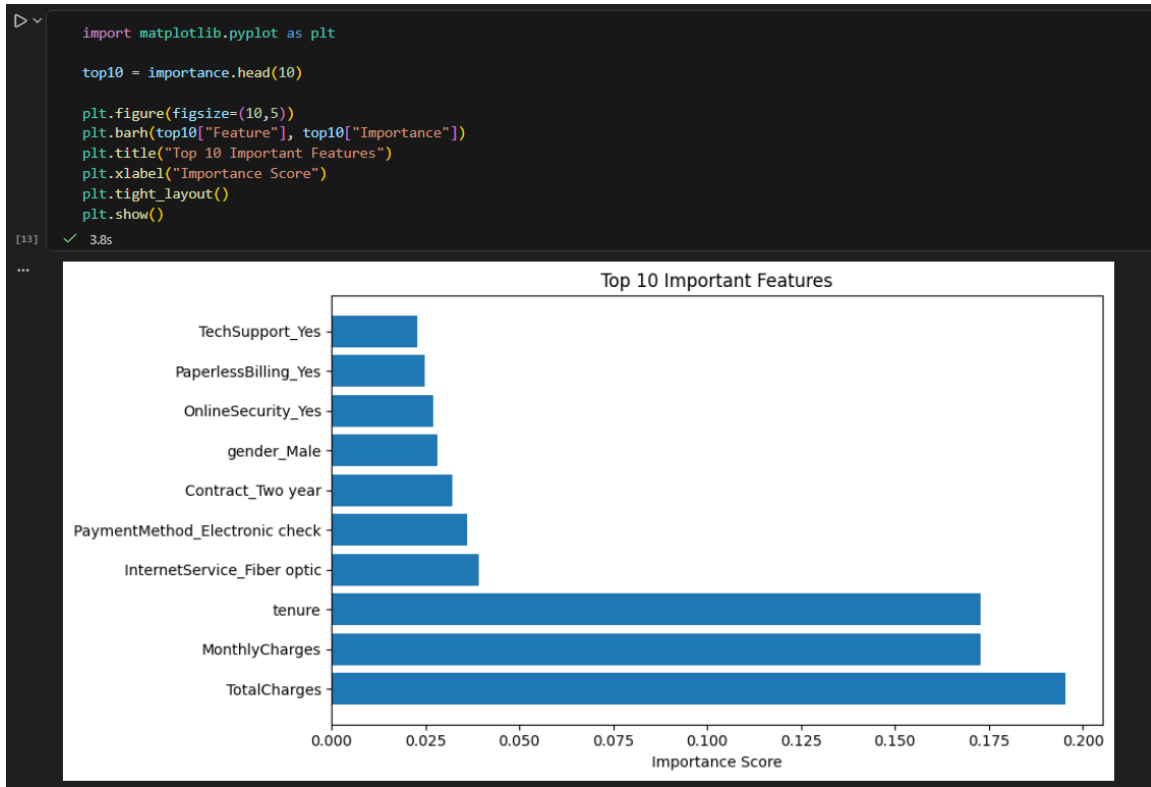
print("Data split files saved successfully.")
```

[8] ✓ 0.1s

```
... Data split files saved successfully.
```

TASK 3 – FEATURE SELECTION

Random Forest Feature Importance identified key features including Tenure, Monthly Charges, Total Charges, Contract Type, Tech Support, and Online Security.



TASK 4 – MODEL SELECTION

Logistic Regression and Random Forest were considered. Random Forest was selected due to its superior classification performance and robustness.

```
lr_model = LogisticRegression(max_iter=1000)

rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

print("Models created successfully")
```

[2] ✓ 0.0s

... Models created successfully

```
print("Model 1: Logistic Regression")
print("Model 2: Random Forest Classifier")
```

[3] ✓ 0.0s

... Model 1: Logistic Regression
Model 2: Random Forest Classifier

TASK 5 – MODEL TRAINING

The Random Forest model was trained using 100 estimators and random_state=42.

```
▶ 
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

print("Model Trained Successfully")
```

[4] ✓ 1.6s

... Model Trained Successfully

```
▶ 
joblib.dump(
    model,
    r"C:\Users\dell\OneDrive\Desktop\Customer_Churn_Project\Outputs\churn_model.pkl"
)

print("Model Saved Successfully")
```

[5] ✓ 0.1s

... Model Saved Successfully

TASK 6 – MODEL EVALUATION

Accuracy: 79.21%

Precision: 65%

Recall: 46%

F1 Score: 54%

ROC-AUC Score: 83.44%

```
print("Accuracy:", accuracy_score(y_test, pred))
print("Precision:", precision_score(y_test, pred))
print("Recall:", recall_score(y_test, pred))
print("F1 Score:", f1_score(y_test, pred))
```

[10] ✓ 0.0s

... Accuracy: 0.7920511000709723
Precision: 0.6526717557251909
Recall: 0.4584450402144772
F1 Score: 0.5385826771653544

```
from sklearn.metrics import roc_auc_score

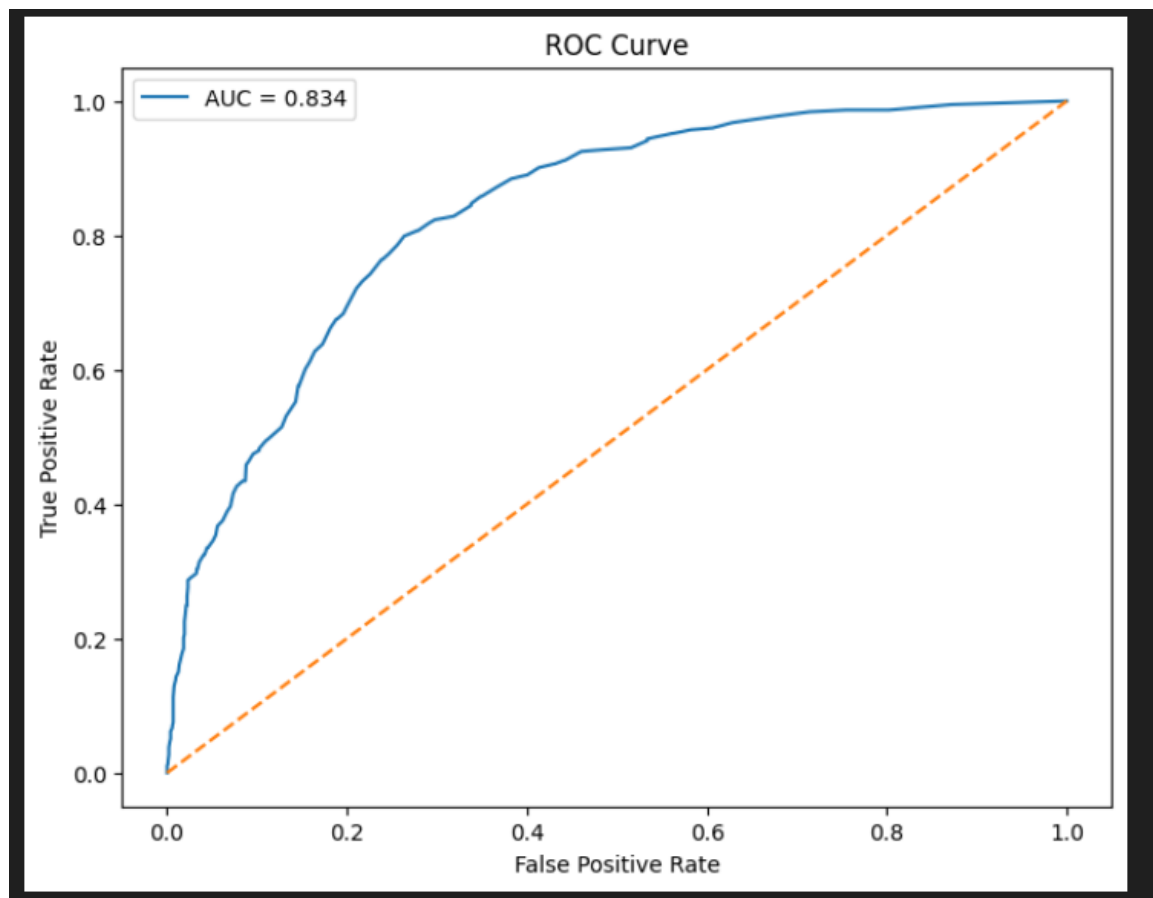
# Probability predictions
y_prob = model.predict_proba(X_test)[:, 1]

roc_auc = roc_auc_score(y_test, y_prob)

print("ROC-AUC Score:", roc_auc)
```

[11] ✓ 0.3s

... ROC-AUC Score: 0.8344361692216922



```
print("\nClassification Report:")
print(classification_report(y_test, pred))
```

[13] ✓ 0.0s

...

Classification Report:				
	precision	recall	f1-score	support
False	0.82	0.91	0.87	1036
True	0.65	0.46	0.54	373
accuracy			0.79	1409
macro avg	0.74	0.69	0.70	1409
weighted avg	0.78	0.79	0.78	1409

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, pred))
```

[15] ✓ 0.0s

...

Confusion Matrix:

```
[[945  91]
 [202 171]]
```

RESULTS AND FINDINGS

Customers with shorter tenure, month-to-month contracts, higher monthly charges, and lack of technical support showed higher churn probability.

CONCLUSION

The Random Forest model successfully predicted customer churn with 79.21% accuracy and can support customer retention strategies.

FUTURE SCOPE

Hyperparameter tuning, XGBoost implementation, deployment as a web application, and real-time churn monitoring.

REFERENCES

Scikit-Learn Documentation
Pandas Documentation
Telco Customer Churn Dataset